

Applications of Nanomaterials in Agricultural Production



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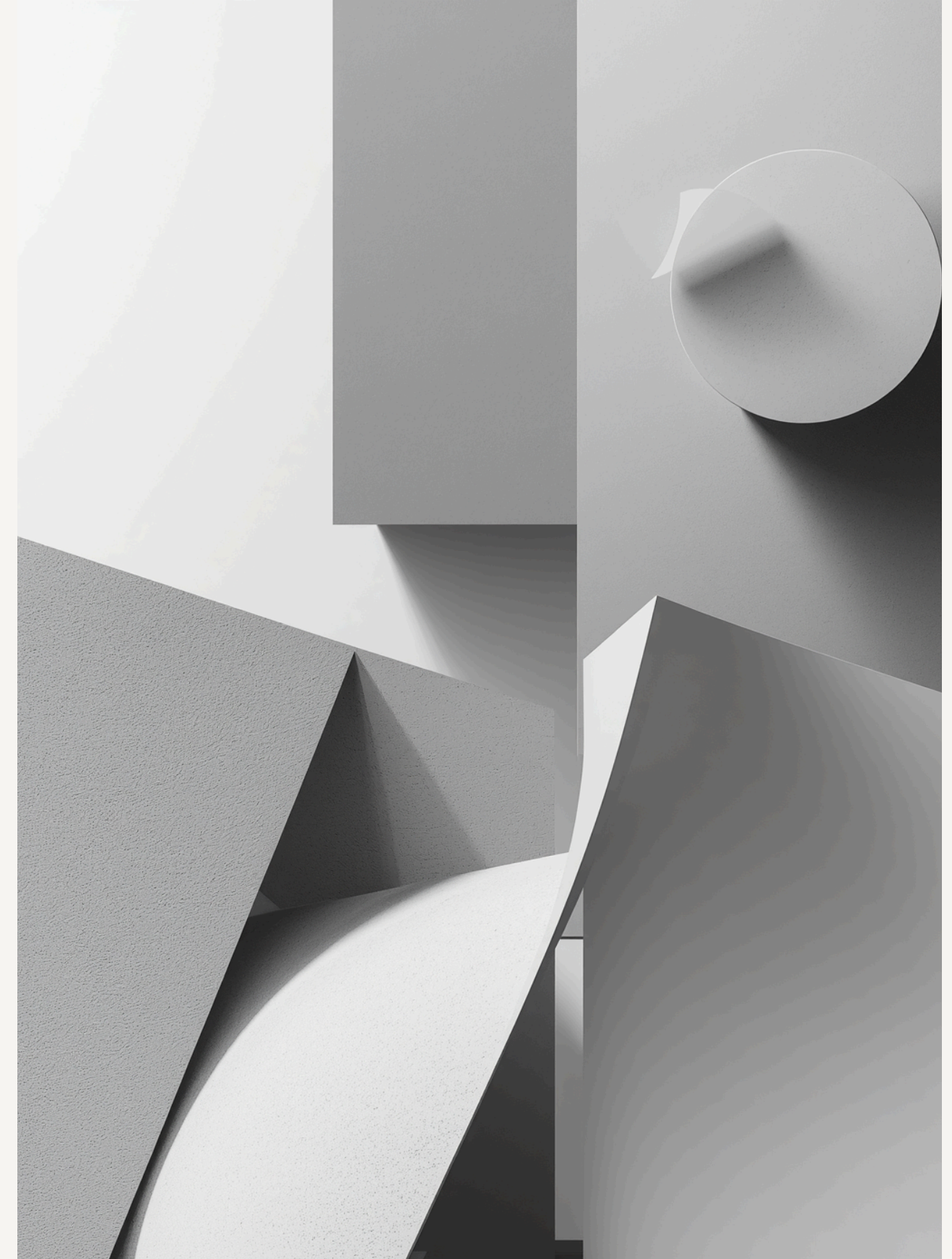
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Keywords

Core Concepts

These keywords represent the foundational themes explored in nanomaterial applications for agricultural advancement.

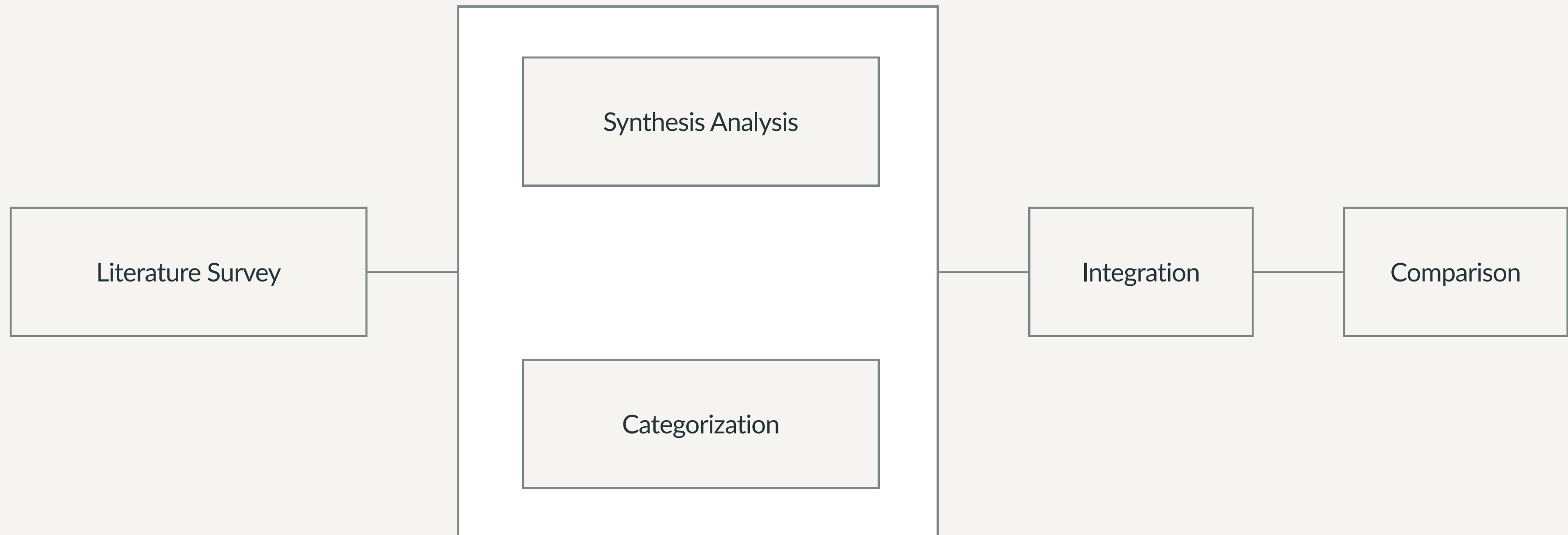
- ▶ Nanotechnology / Nanomaterials: Materials at nanoscale with unique properties; various synthesis methods including physical, chemical, and green approaches.
- ▶ Nano-fertilizers: Improve nutrient uptake efficiency and crop yield through targeted delivery systems.
- ▶ Nano-pesticides: Enhance pesticide delivery effectiveness while improving environmental safety profiles.
- ▶ Nano-sensors: Enable detection of pesticides and ethylene gas, supporting precision agriculture practices.
- ▶ Sustainable Agriculture & Stress Resistance: Eco-friendly practices that help plants resist abiotic and biotic stresses while enhancing growth.

Objectives of the Research



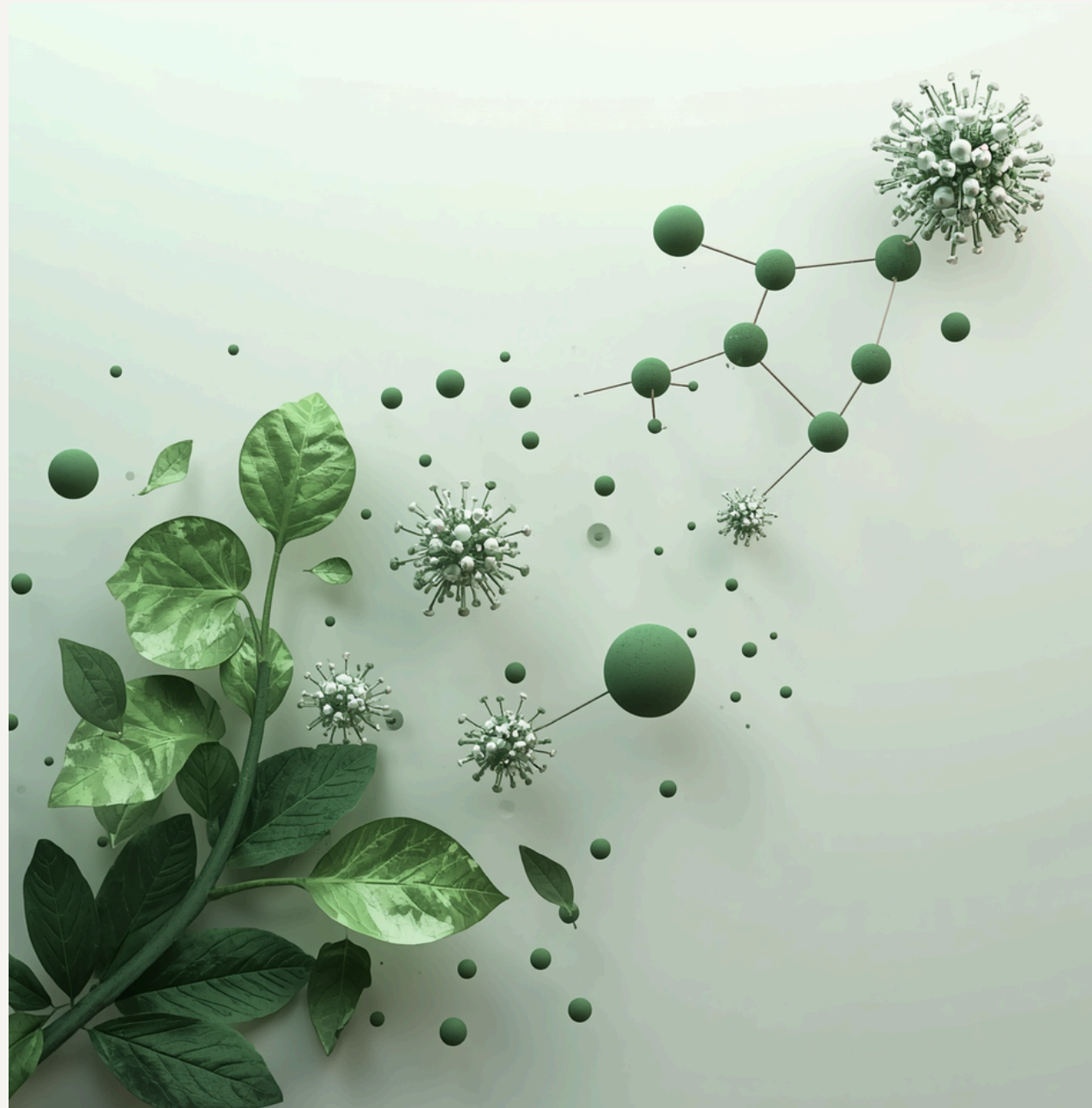
- 01 Review and summarize recent advances in nanomaterial applications in agriculture, including synthesis methods and emerging technologies.
- 02 Analyze applications in transgenics, product preservation, and pesticide efficiency to understand current implementation strategies.
- 03 Analyze applications in stress resistance and growth enhancement, examining how nanomaterials help plants resist abiotic and biotic stresses.
- 04 Identify challenges and propose safe, efficient use strategies for nanomaterials to support sustainable agricultural development.

Methodology Overview



Results – Green Synthesis and Product Preservation

Green synthesis using plant extracts (mango peel, banana peel, tea leaves) offers a cost-effective and eco-friendly approach to nanomaterial production. Nanocoatings such as chitosan/nano-silica and chitosan/nano-TiO₂, along with nanopackaging incorporating nano-Ag and nano-TiO₂, significantly extend fruit shelf life and reduce spoilage.



Results – Pesticide Efficiency and Plant Growth

01

Pesticide Efficiency

Nanomaterials significantly improve pesticide delivery efficiency and enable rapid residue detection through advanced nano-sensors utilizing gold/silver nanoparticles and carbon nanotubes for precision monitoring.



02

Plant Growth

Enhanced seed germination, photosynthesis, and nutrient absorption are achieved through nanomaterials. ZnO nanoparticles notably improve chlorophyll content and photosynthetic efficiency in plants.

Results – Stress Resistance and Transgenic Applications

01



Nanomaterials effectively mitigate abiotic stresses including drought, salinity, extreme temperatures, and heavy metal toxicity by regulating cellular balance and protective mechanisms.

02



Combat biotic stresses from pathogens and pests through antioxidant regulation, enhancing plant defense systems and improving overall crop resilience.

03



Emerging nanomaterial-based gene delivery systems using carbon nanotubes, gold nanoclusters, and mesoporous silica nanoparticles enable efficient, non-toxic plant transformation.

Future Prospects and Challenges



- 01 Integration with AI and IoT enables smart, precision agriculture through real-time monitoring, automated responses, and data-driven decision making for optimized crop management.
- 02 Comprehensive safety assessments are needed to evaluate nanomaterial impacts on human health, ecosystems, and food safety before large-scale agricultural implementation.
- 03 Cost reduction strategies are essential for widespread adoption. Current nanomaterial production costs must decrease through scalable manufacturing and efficient synthesis methods.
- 04 Regulatory frameworks must be developed to govern nanomaterial use in agriculture, ensuring standardized testing protocols and clear guidelines for safe, responsible adoption.

Thank You

2025



Nanotechnology offers transformative potential for sustainable agriculture and food security.

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Questions and discussion welcome